P3: 3.1- Contapache Secure Configuration of Web Server

Ritu Pujari

Docker Pull Command: docker pull ritupujari/contapache:2.0

Table of Contents:

Part 1.1: Run Apache server into the container and map it to a host machine port 2
Part 1.2 2
Part 3: Start a secure Apache HTTPS in the container and map it 3
Part 4: 4

Part 1.1: Run Apache server into the container and map it to a host machine port

In this part, the httpd Docker image was pulled from Docker Hub and ran as a container named ‘contapache.’The host machine’s port 8090 was mapped to the container’s port 80. This allowed access from outside to the Apache web server using localhost:8090

The screenshot below shows the successful output of accessing the container web server using curl. The default Apache page (“It works!”) verifies that the container is running correctly and the Apache service is running inside the container.

```
ritupujari@cy5001server:~$ curl localhost:8090
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<title>It works! Apache httpd</title>
</head>
<body>
<p>It works!</p>
</body>
</html>
ritupujari@cy5001server:~$ _
```

Figure 1.1: Apache default page

Part 1.2

The original index.html file found in the container was replaced by a custom index.html file that contained my name and date. The new index.html file was created on the host Linux machine and copied into the container to /usr/local/apache2/htdocs/. I used the curl command validated the file had been replaced correctly.

The screenshot below shows the updated web page and it also confirms that the index.html was updated successfully in the container. I then committed the container to a new image ritupujari/contapache:1.0 to preserve this change.

```
ritupujari@cy5001server:~$ sudo docker cp index.html contapache:/usr/local/apache2/htdocs/index.html
Successfully copied 2.05kB to contapache:/usr/local/apache2/htdocs/index.html
ritupujari@cy5001server:~$ curl localhost:8090
<html>
<header><title>CY5001 Assignment></title></header>
<body>
I am Ritu Pujari and this is my CY5001 assignment on 26 Feb 2026
</body>
</html>
ritupujari@cy5001server:~$
```

Figure 2: Custom index.html deployed

Part 3: Start a secure Apache HTTPS in the container and map it

To set up HTTPS, OpenSSL was installed in the container, and a self–signed SSL certificate with a subject of C=US, ST=MA, L=NEU, O=MSCY, CN=pujari.ri.cy5001.com was created. This certificate and private key were saved to /usr/local/apache2/conf/ssl/. To enable SSL, I created a Virtual Host in the httpd-ssl.conf file on Port 443, turned SSLEngine on, and specified the SSL Certificate and private key file. The include directive to load the httpd–ssl.conf file was enabled. After restarting, Apache the above configuration changes were in effect enabling my Web Server to run securely using HTTPS in addition to standard HTTP.

After configuring HTTPS, a new image of the container was created and tagged as ritupujari/contapache:2.0, creating a secure copy of the image while retaining the original ritupujari/contapache:1.0, the screenshot confirms that both the versions were successfully created.

```
ritupujari@cy5001server:~$ sudo docker image ls
REPOSITORY              TAG       IMAGE ID       CREATED
 SIZE
ritupujari/contapache   2.0       7be93d882f49   6 minutes ago
 138MB
ritupujari/contapache   1.0       50c7a3446113   5 hours ago
 138MB
<none>                  <none>    b9453e077627   5 hours ago
 117MB
httpd                   latest    10e58ed1b203   2 days ago
 117MB
hello-world             latest    1b44b5a3e06a   6 months ago
 10.1kB
```

Figure 3: Docker images committed

Part 4:

After committing the SSL-enabled image as tag 2.0, the previous container had been stopped and removed, and a new container based on the ritupujari/contapache:2.0 image was created. In the new container, The host port 8090 was mapped to the container port 80 and the host port 8443 to container port 443. Both the HTTP on port 8090 and HTTPS on port 8443 returned an index.html page, confirming that both protocols are functioning correctly.

Finally, the docker login command was run using the user ID ritupujari to log into Docker Hub to authenticate. Both versions (1.0 and 2.0) of the image were pushed to the Docker Hub repository, ritupujari/contapache. The screenshot below shows terminal window output of this process. You can see the docker login was successful and the two images were successfully uploaded to Docker Hub.

```
ritupujari@cy5001server:~$ sudo docker login
Authenticating with existing credentials... [Username: ritupujari]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
ritupujari@cy5001server:~$ sudo docker push ritupujari/contapache:2.0
The push refers to repository [docker.io/ritupujari/contapache]
deb37d4323f4: Pushed
884efe8beadc: Pushed
7618dcbb5d95: Mounted from library/httpd
99fe004f8a12: Mounted from library/httpd
1e5c482a53a1: Mounted from library/httpd
5f70bf18a086: Mounted from library/httpd
e348b0b65a27: Mounted from library/httpd
a257f20c716c: Mounted from library/httpd
2.0: digest: sha256:16526360f79f55060854e60a91270cca92c1bfae839ad69613d066ab8c2b938e size: 1991
ritupujari@cy5001server:~$ sudo docker push ritupujari/contapache:1.0
The push refers to repository [docker.io/ritupujari/contapache]
fa5d0bcee6a7: Pushed
7618dcbb5d95: Layer already exists
99fe004f8a12: Layer already exists
1e5c482a53a1: Layer already exists
5f70bf18a086: Layer already exists
e348b0b65a27: Layer already exists
a257f20c716c: Layer already exists
1.0: digest: sha256:e5e96610c64e2145fabcc83e1b768148f299b090d199c03119e1c34b971dfc09 size: 1784
```

Figure 4.1: Docker login and Push to Docker Hub

The snapshot captured confirms the successful retrieval of the ritupujari/contapache:2.0 image from Docker Hub, demonstrating that this image can be publicly accessed. Both the curl command to the HTTP port and HTTPS port return the same custom index.html page, with name and date on it, indicating that any modifications made during the assignment can be verified.

```
ritupujari@cy5001server:~$ sudo docker pull ritupujari/contapache:2.0
2.0: Pulling from ritupujari/contapache
Digest: sha256:16526360f79f55060854e60a91270cca92c1bfae839ad69613d066ab8c2b938e
Status: Image is up to date for ritupujari/contapache:2.0
docker.io/ritupujari/contapache:2.0
ritupujari@cy5001server:~$ sudo docker run -dit --name contapache -p 8090:80 -p 8443:443 ritupujari/contapache:2.0
287ed4d007d17283ca5c74028de8ea52218f248ac5f40374f8b1491bcd90b9d7
ritupujari@cy5001server:~$ curl localhost:8090
<html>
<header><title>CY5001 Assignment></title></header>
<body>
I am Ritu Pujari and this is my CY5001 assignment on 26 Feb 2026
</body>
</html>
ritupujari@cy5001server:~$ curl -k https://localhost:8443
<html>
<header><title>CY5001 Assignment></title></header>
<body>
I am Ritu Pujari and this is my CY5001 assignment on 26 Feb 2026
</body>
</html>
ritupujari@cy5001server:~$
```

Figure 4.2: Docker Pull and changes verified

The final screenshot confirms that both tags (1.0 and 2.0) of the ritupujari/contapache Docker Hub repository are publicly available and accessible; and can be retrieved via the docker pull command provided at top of this report.

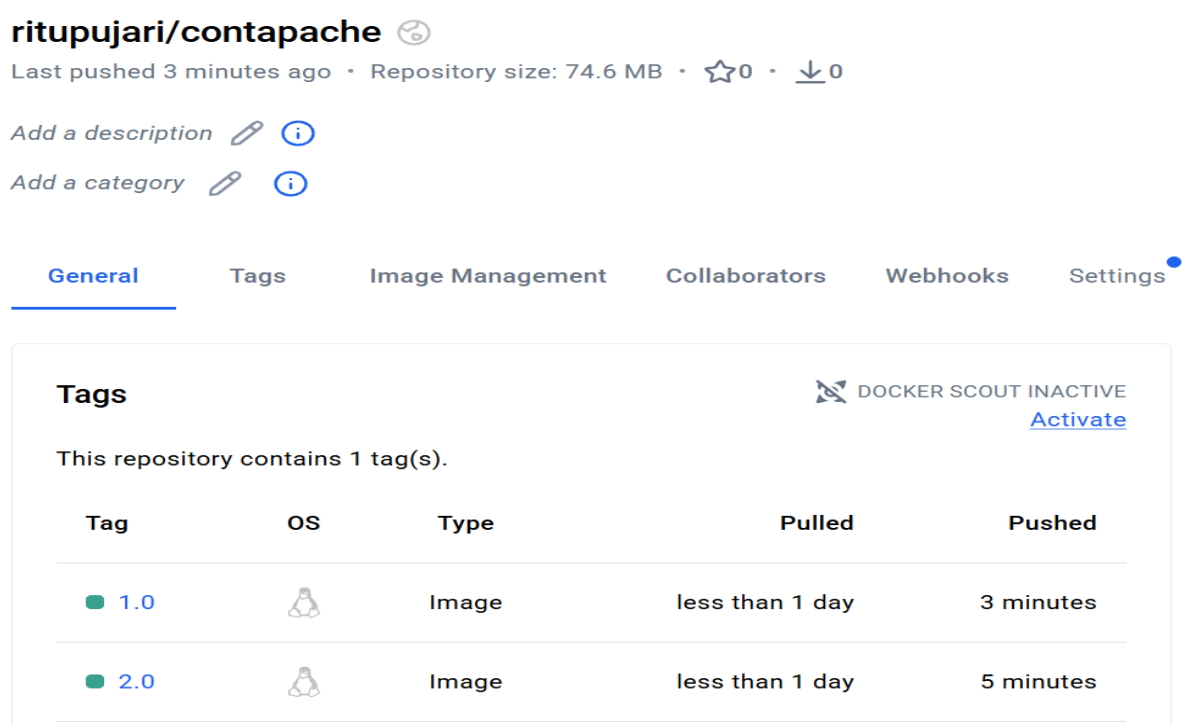

Figure 4.3: Docker Hub Repository